

You will need the following to install minikube on the Ubuntu operating system.

Container or virtual machine manager	Architectures
Docker, QEMU, Hyperkit, Hyper-V, KVM, Parallels, Podman, VirtualBox, or VMware Fusion/Workstation	2 CPUs or more 2GB of free memory 20GB of free disk space

Go to *Settings* and click on the *About* tab to verify the Ubuntu version.

Next we will install minikube on the Ubuntu operating system.

Minikube installation on Ubuntu 22.04.1 LTS

To install the latest minikube stable release on Ubuntu 22.04.1 LTS execute the following commands:

```
osfy@ubuntu-22-04-1-lts:~/Desktop$ curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube_latest_amd64.deb
```

```
% Total    % Received % Xferd Average Speed   Time
Time      Time      Current
```

```
      Dload  Upload    Total   Spent    Left  Speed
100 26.1M  100 26.1M    0     0  1430k      0  0:00:18
0:00:18 --:--:-- 1231k
```

```
osfy@ubuntu-22-04-1-lts:~/Desktop$ sudo dpkg -i minikube_latest_amd64.deb
```

```
Selecting previously unselected package minikube.
(Reading database ... 204624 files and directories currently installed.)
```

```
Preparing to unpack minikube_latest_amd64.deb ...
Unpacking minikube (1.28.0-0) ...
Setting up minikube (1.28.0-0) ...
```

Start your minikube cluster. If you face any issues, stop the cluster, delete it and execute the *minikube start* command again.

Now verify the existing status of minikube by using the *minikube status* command:

```
osfy@ubuntu-22-04-1-lts:~/Desktop$ minikube status
```

We have installed minikube and verified it. The next step is to verify the Java and Maven installation, create a Docker image, and deploy it using Kubernetes YAML in the minikube cluster.

```
osfy@ubuntu-22-04-1-lts:~/Desktop$ minikube stop
🛑 Stopping node "minikube" ...
🔴 Powering off "minikube" via SSH ...
🔴 1 node stopped.
osfy@ubuntu-22-04-1-lts:~/Desktop$ minikube delete
🔥 Deleting "minikube" in docker ...
🔥 Deleting container "minikube" ...
🔥 Removing /home/osfy/.minikube/machines/minikube ...
💀 Removed all traces of the "minikube" cluster.
osfy@ubuntu-22-04-1-lts:~/Desktop$ minikube start
😄 minikube v1.28.0 on Ubuntu 22.04
🌟 Automatically selected the docker driver. Other choices: none, ssh
🔧 Using Docker driver with root privileges
👍 Starting control plane node minikube in cluster minikube
📡 Pulling base image ...
🔥 Creating docker container (CPUs=2, Memory=7900MB) ...
📡 Preparing Kubernetes v1.25.3 on Docker 20.10.20 ...
   ▪ Generating certificates and keys ...
   ▪ Booting up control plane ...
   ▪ Configuring RBAC rules ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: storage-provisioner, default-storageclass
👍 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
osfy@ubuntu-22-04-1-lts:~/Desktop$
```

Figure 1: Start minikube cluster